closely, for the reason possibly that it has been standard for a number of years.

LUBRICATION OF LOCOMOTIVE CYLINDERS.

The committee compiled the following questions, which were submitted to members:

1. Have you experienced difficulty in lubricating modern passenger and freight locomotives, with steam pressures from 180 lbs. to 230 lbs., with the up-to-date lubricators which operate by condensation of steam, adjustable feed, etc., and, if so, state experience and causes for same?

2. What recent improvements have been made in the hydrostatic-feed lubricators, to add to their efficiency and economical operation? Give explanation, experience and results.

3. If familiar with any device, independent of the hydrostatic-feed lubricators as furnished by the manufacturers, which increases their efficiency or economical use, give description, principle of operation, etc.

4. For the lubrication of valves and cylinders on modern locomotives, state your objections, if any, to the up-to-date hydrostatic-feed lubricator

5. What types of mechanically operated lubricators have you used? State method of application, principle of operation, control and regulation of feed, location of device on the locomotive, with experience and results.



C. H. Rae.

Chairman, Committee on Lubrication of Locomotive Cylinders.

6. For the lubrication of modern locomotives, valves and cylinders, state your objections, if any, to the mechanically operated lubricator.

7. Are you familiar with any method, or device, for admitting the oil into the steam before it reaches the saddle casting? If so, state details, experience and results.

8. Have you had any experience with the oil-delivery pipe attached to deliver the oil into the steam-way in the saddle casting, instead of at the steam chest or valve chamber? State results.

9. What has been your experience in the lubrication of locomotive cylinders where superheated steam has been used? In this connection state the degrees of temperature acquired, if hydrostatic or mechanically operated lubricators were used, and the results from both, where both have been used.

10. Do you consider it necessary to lubricate cylinders of locomotives using superheated steam, independently of the valves, using any type of lubricator?

11. What is your experience with the lubrication of cylinders on Mallet compound locomotives, using superheated steam with the hydrostatic-feed lubricator? With the mechanically operated lubricator?

12. Do you consider it necessary to lubricate the valves and cylinders on the high and low pressure cylinders of Mallet compound locomotive (by lubricators of whatever type) piped independently to each valve chamber and cylinder?

The replies to question No. 1 were universal in expression, that there is no serious difficulty now experienced in lubricating modern passenger and freight locomotives with the up-to-date lubricators of the hydrostatic-feed type. Several of the members, in reply to question No. 2, have referred to a recent improvement in the hydrostatic lubricator, whereby it is made possible to close or shut off feeds without interference with the feed-valve adjustments. This is an economical feature, but of such recent date that no comparative data have been submitted as to the results obtained. The replies to question No. 3 were limited. One member replies as follows: "The automatic steam-chest choke plugs, which feed against a constant boiler pressure in the oil pipe, and not against a pressure that fluctuates with pressure in steam chest, will result in a high degree of efficiency in the lubricator."

All replies to question No. 4 indicated there were no objections. The replies to questions Nos. 5 and 6 embody the same general information, but were limited in number. Some are given herewith:

a. "Have used several types of mechanically operated forced-feed lubricators. Three of those experimented with were driven from a moving part of the engine. This was unsatisfactory, since the feed is required that does not vary with the speed of the engine. More oil is required per minute while the engine is worked hard at low speeds than when running fast with a light throttle. To overcome this we used some forced-feed lubricators with independent air motor. None of them, however, have been satisfactory. It is difficult to apply to the filling holes screens that are fine enough to keep out dirt, without restricting the speed at which the oil can be put in, and the enginemen pull them out. The consequence is that the plungers wear rapidly, and after a time the feed becomes irregular. They are troublesome to maintain, and under conditions specified, in answer to question No. 1, gave no better results than the hydrostatic lubricator."

b. "We have experimented with two or three makes of mechanical-feed lubricators, but have been unsuccessful in getting as good results as we do with the hydrostatic-feed lubricators. The trouble with the mechanical-feed lubricator seems to be due to the difference in temperature; in warm weather, or during the warm part of the day, they would feed sufficient oil, but in cold weather, or at night when the weather is colder, they do not feed as well as when it is warmer."

c. "We have used two mechanical lubricating devices, which received their motion from a connection made to the valve stem and oil chamber, and the regulating devices were located in the cab. Our test record shows that after considerable experimenting the device operated fairly well, but after a time it gave considerable trouble and was finally removed on account of not giving satisfactory results. Mechanical means are objectionable, in that the device is so complex, and consists of so many parts, it is difficult to keep joints from leaking. To be properly installed, all pipes must be kept constantly filled and under pressure, with return valves at distributing points. On account of mechanical movements, parts will wear, and it will be more expensive to maintain, and in making repairs more oil is lost than with hydrostatic lubricators. Devices of this sort do not appear to have as yet been perfected to such an extent as to make them thoroughly reliable."

d. "Have used a force-feed lubricator with pipe connections to all the driving boxes and to the steam chest. Operated by a mechanism deriving its motion from a connection to the Walschaert link. Lubricator was located in the cab. While reducing friction on the journal bearings; and delivering oil to the steam chest, I do not consider it entirely satisfactory, for the reason that there is nothing to indicate that the pumps are working properly and delivering the oil until the valves become dry."

e. "Our objection to mechanically operated lubricators is that they are necessarily more complicated than the hydrostatic lubricator, without showing any beneficial results."

f. "Have tried some, but have not obtained satisfactory results."

g. "We have in the past used on some of our two-cylinder compound engines mechanically operated lubricators. Our objection to it was the rapid wear of the parts, leakages and annoyances."

h. "Using displacement plunger force feed. Secured in suitable location in cab, and operated by a series of rods and bell cranks, which are connected, either to eccentric blades and independent eccentric on rear or other convenient axle, or by reducing-arm on back crank pin. Device subject to varying control, and fed by increasing or decreasing

stroke of plungers by means of adjustment thumb-nut. Results as to economy and distribution very satisfactory. Slight trouble experienced in experimental stage, due to method of operating lubricator."

There was but one reply to question No. 7, citing experience which is herewith submitted:

"Present method in use with mechanically operated lubricator is to tap direct through dome into steam pipe, or throttle box. This method has proven to be superior to the old method of admitting to steam chest direct."

In this connection, and pertinent to the question, a member refers to a test:

"Delivering the oil from the right side of the lubricator into the left cylinder, passing it through the smoke box, and from the left side of lubricator into the right cylinder in the same manner. The gases in the front end superheating the oil, and evaporating the water of condensation, putting the oil into the cylinders superheated."

The substance of the replies to question No. 8 from those who have had experience was in accord with the report from a member who has tried this method extensively:

"On slide-valve engines consider results distinctly superior; on piston-valve engines reports vary, but general experience is that better results are obtained by putting oil into the steam passage."

Questions Nos. 9 and 10 have been considered jointly by the members. In the information submitted, the important features of this phase of the subject are well expressed in the quotations, as follows:

a. "Superheat varies from 550 to 580 deg. F. Our regular practice is to use one feed pipe to valve chest, applying it to go to each end of valve in center of the valve bushing. This lubricates the valve very well, but it is questionable whether delivering oil to the steam passage is not preferable. We also use one feed to each side of the cylinder, at the top and in the center. This is not used unless required. When working with full throttle and long cut-offs, over considerable distances, the cylinder feed was found necessary. There is no difficulty whatever in obtaining satisfactory lubrication in superheater engines with hydrostatic lubricators, on account of difference of pressure between steam chest and boiler, previously mentioned. In fact, there is considerably less difficulty than on engines using saturated steam. In spite of this the piston-ring wear is far more rapid."

b. "On our superheaters we carry 170 lbs. boiler pressure; the maximum degree of superheat is 225. On a number of superheater engines we introduced an oil both at the steam cavity and into the cylinder direct; the latter connection was found unnecessary. The only trouble which we had with superheater engines was to get our men to use the drifting valve when the engine was shut off. If this is not done, trouble was experienced with the bushings cutting out, but on districts where the drifting valve is put on we are having very good success."

All replies to question No. 11 pertain to the satisfactory use of the hydrostatic lubricator. No information was furnished concerning the use of mechanically operated lubricators on Mallet compound locomotives using superheated steam.

Replies to question No. 12 were at variance as to the location of the oil pipes, but it was the consensus of opinion that piping to the high-pressure valves and cylinders only was insufficient for satisfactory lubrication.

The experience of the committee, supplemented by the information received from the members of the association, warrants the assertion that there is no serious difficulty now experienced with the use of the hydrostatic-feed lubricator. The recent addition of a stop-feed feature is an improvement, rather tending to economical operation than to efficiency, and it behooves the manufacturers of this device to keep pace with, or in advance of, the constantly increasing demands.

The information obtained from the members, and cited in the foregoing, confirms the experience and opinion of the committee, that a properly constructed hydrostatic lubricator meets the locomotive requirements better than a mechanically operated lubricator, for the several reasons:

1. Familiarity with care and operation by the different classes of labor whose duties are in connection with its use.

2. Simplicity of design and substantial in construction; the operating parts being better protected from disarrangement or breakage.

3. A more accurate regulation of the amount of oil applied to the valves and cylinders under the varying conditions of service performed by the locomotive at different speeds and points of cut-off.

4. Because of less complication in construction and attachments, a corresponding less expense of maintenance.

The more general custom of delivering the oil to the steam chest or valve chamber is open to question, and there has been some very conclusive evidence submitted favoring the delivery of the oil into the steam at a point where it may become highly attenuated and intermingled with the steam. The presumed effect of extreme temperatures, due to high pressure and superheat upon the oil, has been an objection to delivering the oil in the steam before it reaches the cylinder saddle. Information has been furnished and confirmed by the experience of some of the members, that the efficiency of a properly compounded mineral cylinder oil is not seriously impaired when protected by the steam.

As the reports on this particular feature of the subject are indefinite, the committee recommends further consideration and experimentation. The information obtained from the members who have had the most general and extended experience with locomotives using superheated steam confirms the experience and opinion of the committee, i. e., that the same reasons advanced for the endorsement of the hydrostatic-feed lubricator on locomotives using saturated steam apply to locomotives using superheated steam. The information submitted and quoted in the replies, pertaining to the proper location and number of oil pipes on superheated locomotives, is of much value. The experience thus far is not sufficiently conclusive to justify a decisive recommendation at this time.

However, it is recommended that liberal openings be given to drifting valves, and attention to their proper manipulation, that the temperature of the cylinders may be promptly reduced within the lubricating possibilities of the oil when exposed to the atmosphere.

The problem of satisfactorily lubricating the Mallet compound locomotive is still in process of solution. At present it seems essential to pipe independently to the high and low pressure cylinders. However, the committee has been advised that there has been some experience, with satisfactory results, by eliminating the pipes to the low-pressure valves and cylinder, substituting an auxiliary oil pipe to the receiver with the high-pressure steam connection. This carries sufficient oil over to the low-pressure cylinders to insure good service. Increased efficiency and reduced expense of operation confront the mechanical departments of our railways to a greater extent than ever before. The final and most important recommendation of the committee is a continued live interest in the lubrication of locomotive cylinders.

The report is signed by:—C. H. Rae (L. & N.), chairman; Jos. Chidley (L. S. & M. S.), T. R. Cook (Penn.), J. F. DeVoy (C. M. & St. P.), J. F. Walsh (C. & O.), S. T. Park (C. & E. I.) and P. M. Hammett (M. C.).

DISCUSSION ON LUBRICATION FOR LOCOMOTIVE CYLINDERS.

George L. Fowler (*Railway Age Gazette*): I will simply repeat some information that I obtained the other day from Mr. Muzzey, who is mechanical engineer for the Long Island. They have been experimenting for the last six or eight months with a mechanical lubricator for cylinders, in which a mixture of oil and graphite is pumped into the cylinder, operated mechanically, and after a run of about two hundred miles the faces of the steam chests, valve seats and cylinders become coated with this film of graphite, and as far as that one engine is concerned, the operation has been very successful and very satisfactory, and they are inclined to think that it will reduce the internal friction of the engine to such an extent as to produce some saving in fuel.

There has been no data published, but it has been so successful that they are about to make an extensive trial, I understand, of some eight or ten engines, in which very exhaustive tests will be made to determine exactly what can be done from both a mechanical and economical standpoint in developing this method of feeding.

As the lubricator is so constructed that the old trouble of the graphite settling and becoming a sediment in the bottom of the oil is done away with by the mixture, which keeps it stirred up until it is actually deposited in the steam chest. Hence the operation is wholly satisfactory.

E. A. Miller (N. Y. C. & St. L.): We have found that very good results can be obtained by applying the graphite at the commencement of each trip to a cup, separate from the lubricator, and applying this graphite cup to our steam chests in addition to the regular lubricator.

George A. Hancock (St. L. & S. F.): We have fifty engines, with the steam temperature about 535 to 565 deg. We tried the mechanically operated lubricator and found it unsatisfactory. We had trouble on our connections with the

piece of rubber hose. Steam from the air-pump exhaust pipe passing through a coil placed within the oil cup maintains the oil in a fluid state. The Chicago, Rock Island & Pacific is using a lubricator shown in Fig. 6 on two Pacific type locomotives. The oil receptacle stands upon the running board over the right cylinder. Exhaust steam passes through a pipe leading from the exhaust cavity in the cylinder casting to the oil cup. The accumulation of condensation in the oil cup lifts the lubricant into the delivery pipe, exhaust steam carrying it to the flange. The flow of oil is controlled by the engineer through a globe valve operated by a rod extending back to the cab. A check valve placed in the pipe leading from the cylinder exhaust cavity prevents a back flow of oil through this pipe.

There are two lubricators in use employing lubricant in the solid form: the Collins and the Turnbull. The Collins is illustrated in Fig. 7. A bracket attached to the frame

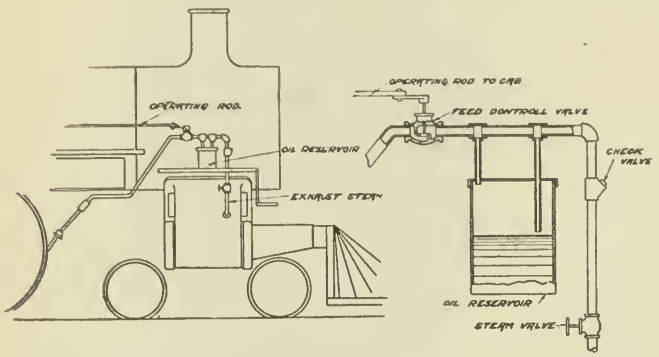


Fig. 6—Flange Oiler; Chicago, Rock Island & Pacific.

supports the lubricator in position before the driver. The angle of the lubricator is adjustable to suit conditions imposed by the location of the bracket, on which it is laterally adjustable. The angle should be as nearly as possible 25 deg. from a line parallel to the axle and it should be placed on the horizontal center line of the wheel. The location recommended by the manufacturers is shown in Fig. 7. The lubricator itself is a tube of rectangular cross-section through which slides the lubricator block of hard grease. Below this tube is a spring-feeding device designed to keep the lubricator block pressed against the flange. Pressure from a coil spring acts through a dog engaged in one of a series of notches on the under side of the lubricator block. As the lubricant becomes worn, the spring is recompressed by means of a downward projecting trigger. At the same time the dog is moved back into engagement with the next notch in the lubricator block. This periodical readjustment is the only attention required to operate the lubricator.

The Turnbull lubricator embodies the same principle, spring pressure holding a cake of hard grease against the flange. The committee is informed by the manufacturer that this device is no longer marketed and a more detailed description will, therefore, be unnecessary. The feed is automatically regulated to suit the varying needs of the flanges. On the Colorado & Southern, however, it has been found that the grease used in the Turnbull lubricator absolutely loses its merit in either wet or cold weather and its use has been abandoned. On the Canadian Pacific, Lines West, hard-greese lubrication has been found unsatisfactory. The committee is unable to state what type of lubricator was used.

Several methods of delivering engine and car oil to the flanges have been tried. The simplest is a piece of pipe or hose secured to the frame in front of the driver in such a manner as to cause the lower end to bear against the flange. The pipe or hose is filled with oil-saturated waste, which acts as a swab, more oil being supplied from time to time. Where the waste comes in contact with the tire, if packed hard, it will glaze and cease to be effective. If loosely packed, it will be drawn out and lost. Crude oil is being used in this manner on the Illinois Central to a limited extent on Pacific type engines with very little apparent benefit. It is not reliable in cold weather.

The Canadian Pacific has experimented with a siphon lubricator using engine oil, but this has proved inefficient and has been abandoned for the fuel-oil lubricator shown in Fig. 5.

Another simple device has been applied to six-wheel switch engines in the Chicago yard of the Santa Fe, and is shown in detail in Fig. 8. It is simply a block of wood resting upon the tire and grooved to fit the flange. A waste-filled cavity contains engine oil which feeds through a ⅜-in. hole to the flange. The block is loosely anchored and is usually applied to forward and back drivers. This method, however, greases the whole tire, which sometimes causes the wheel to slip.

The Chicago, Rock Island & Pacific has 10 Pacific type and 3 consolidation locomotives arranged with piping to carry water from the tank to nozzles spraying against the leading drivers. This means of lubrication is of very limited service. The pipes run horizontally along the engine frames and cause trouble by freezing during winter weather. The Pennsylvania has made use of exhaust steam from the air pump, with what success it is not known. Of the lubricators described, all are in very limited use except those using crude oil.

Flange lubrication on the Santa Fe has been developed to its present state after experiments covering the use of most of the simple devices, such as swabs of oil-saturated waste, water jets operated from the injectors and the block type shown in Fig. 8. These have all proved to be in some manner unsatisfactory. Water and engine oil have both proved too light to satisfactorily resist the action of centrifugal force, being thrown away from the throat of the flange before reaching the rail. Crude oil is now in general use for flange lubrication on the Santa Fe System

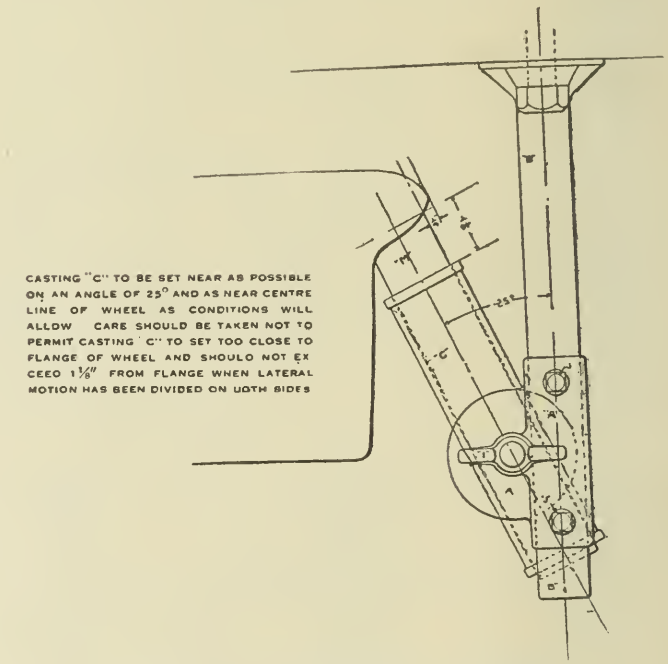


Fig. 7—Hard Grease Flange Lubricator.

Experience indicates that the delivery of a proper lubricant to the flange will reduce the wear of both flange and rail. The committee's information is confined largely to the results obtained by lubrication with crude oil. California crude oil contains a high percentage of petroleum asphalt. When delivered to the rail by the driving-wheel flange, it forms a thin coating of paste on the inside of the ball of the rail which does not run or spread over the top. When all engines on a division are equipped with lubricators, the rails on the outside of curves will become thus coated, and friction will be reduced on all wheels passing over the track. The resulting reduction in flange wear is noticeable on both passenger and freight car wheels, but data are available for locomotive driving wheels and tender truck wheels only. It necessarily follows that train resistance is much reduced on curves.

The block lubricator previously described (Fig. 8), applied to six-wheel switch engines, has extended the period between turning for flange wear in some cases from two months to one year.

On the New Mexico Division, a 65-mile section of which has 288 curves of 6, 8 and 19 deg., it was found impossible to keep engines of the Santa Fe type in service for more than 18,000 to 20,000 miles, but at the present time with lubricators applied they are making as high as 35,000 to 40,000 miles between shoppings with no evidence of flange cutting

On Pacific type locomotives the mileage was increased from between 25,000 and 30,000 to between 75,000 and 100,000. The performance of lubricators has been closely watched in this territory. There are instances where lubrication has not entirely stopped flange wear because of the difficulty experienced in getting engineers to give the lubricators proper attention. On the New Mexico Division, with 100 per cent. of the locomotives equipped, it has been estimated that there has been an increase of about two months in the life of rails, that previously required changing about every thirteen months. This is an increased life of 15 per cent. In territories where no figures are available, it is, however, the opinion of all officials that the flange oilers have materially decreased rail wear. The effect on wear of rails on curves has been no less marked. Where rails on 8-deg. and 10-deg. curves would last only about eight or ten months, conservative figures place the life since general application of flange oilers at about three years.

Flange lubricators on the Southern Pacific have become a necessary part of the locomotive equipment in mountain territory. When not working properly the engineers complain seriously until they are repaired. Locomotives ride easier and smoother around curves without the tendency to climb the rail which is evident when the flanges are run without lubrication.

On the Wabash Pittsburgh Terminal before the application of lubricators the average time between shopping for sharp flanges was eight months. A ten-wheel freight engine, working on a section having maximum curvature of 11 deg. 28 min., would not run three months before developing flange wear to such an extent that tires had to be turned. Since applying the lubricator this engine, working on the same section of the road, was out twenty months before shopping, when tires were turned for tread wear. Since the

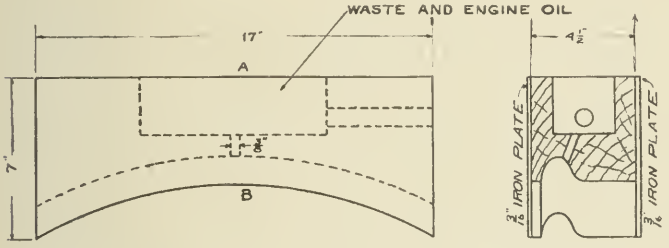


Fig. 8—Flange Lubricator; Atchison, Topeka & Santa Fe.

application to all locomotives there has been no occasion to turn tires for flange wear. It has stopped almost entirely on wheels to which the oil is delivered, and oiling the drivers materially decreases the wear on engine-truck wheel flanges. In some cases, oil has been applied to the trucks where wear was unusually excessive. Mallet locomotives showed some signs of cutting the front flanges on the low-pressure engine. By plugging the nozzles leading to the back drivers on this engine and delivering the same amount of oil that was previously used on both pairs to the front wheels, the cutting has been reduced and it has been found that the other wheels are sufficiently oiled. In addition to the direct benefit to the locomotive, there has been a decrease in the wear of switch points and rails on curves, as well as a decrease in the number of derailments. Flange wear has also been reduced on tender truck wheels.

The Atlanta, Birmingham & Atlantic has gotten as high as 50,000 miles out of locomotives that ran only 13,000 to 15,000 miles before lubrication was adopted. The average length of service for switch engines without lubricators was three or four months at twelve hours per day. Another individual case on this road shows forcibly the effect of lubrication. A ten-wheel locomotive, weighing 130,000 lbs. on drivers, in making 18,958 miles, had cut the front flanges very badly. Owing to the rush of work it was necessary to keep the engine in service if possible. A lubricator was applied and the engine again placed in service. After making 34,495 additional miles, the flanges showed no further wear. The tires were shifted and the engine again put into service. The Elliott system is in use on this railway. The cost of oil has been about 3 cents per 100 engine miles.

A service test of a yard engine was made on the Kansas City Terminal. The yard where this engine worked is so located that the front and back flanges on the left side of the engine became worn to such an extent that the engine began to climb the rail after being in service about three months. It was then necessary to change tires. The engine equipped with a flange oiler was put in service with a new set of tires and remained in service under these conditions over a period of thirteen months, when it was shopped for other repairs. The flanges were still in good condition. The results obtained were made possible by the careful attention the lubricator is reported to have received from the engineer. The foregoing instances indicate the service of flange lubrication under some of the worst conditions affecting flange wear reported to the committee.

The report is signed by:—M. H. Haig (A. T. & S. F.), chairman; T. W. Heintzelman (So. Pac.) and D. J. Redding (P. & L. E.)

After presenting this report Mr. Haig spoke as follows: The locomotives in passenger service upon which flange wear is most prevalent are the Pacific type and the consolidation. However, where types are employed with a longer wheel base than the consolidation these are reported as being subject to the greatest flange wear. In general, if other conditions are equal, it is the type with the longest rigid wheel base on which flange wear is the greatest. There are some notable exceptions to this rule, however, in passenger service. There are two classes of expense due to excessive flange wear. One results from the metal lost in producing standard flanges after they have been badly worn. The other is due to the loss of revenue and the cost of repairs when necessary to turn or remove tires between regular shoppings of the engine. There are four types of lubricators designed to use asphaltum base crude oil. These are the Chicago, the Southern Pacific, the Canadian Pacific and the Rock Island. Where no other name is known, that of the railway upon which the lubricator has been developed is used. Crude oil is now in general use for flange lubrication on the Santa Fe system. There are no data on the Santa Fe to show the exact increase in the life of rails on curves effected by the flange oilers.

DISCUSSION ON FLANGE LUBRICATORS.

T. O. Sechrist (C. N. O. & T.): Before introducing this flange lubricator on our road we would not average over 12,000 miles before we had to change tires, and after the application of the flange lubricator we got as high as 80,000 miles. We have used the Turnbull lubricator. It is made up of two solid cakes, and we could not make a trip, say, of 137 miles before the cakes would be used up. This was caused, of course, by the friction and heat that was developed. Since we have used the Chicago lubricator and asphaltum crude oil as a base, in connection with the lubricator we have gotten over that trouble. I believe that this committee has given us the information that is necessary, for all of the railways that have adopted the flange lubricator will verify the statements made.

F. F. Gaines (Cent. of Ga.): In the past we have been very much troubled with the repeated cutting of flange. It was not due so much to curvature of the line, either. About a year and a half ago we started in experimenting by putting on a few engines the Southern Pacific type of lubricator, and we got good results so long as we followed up the engineer, but the trouble was that he did not use it always. It was a little out of the way and not handy for him to use it. With some engineers we got excellent results. We then tried the Ohio lubricator. It seems with the use of that lubricator the engineer can watch the feeding of the oil, and it is conveniently handy for him, and he is more apt to use it. Our results with it have been very good, indeed. If you take the Pacific type of engines, where we turned flanges about three times on an average we now have every reason to believe that we are going to run them right through from engine repairs to engine repairs without having to turn tires. On the consolidation engines the results are equally favorable. We have also tried the stick lubricators, but my experience with them is the same as that of Mr. Sechrist. They wear out too fast. Taking it all in all, it seems to me that some lubricator of the form of the Ohio, with crude asphaltum as a base, is the best for the purpose.

E. W. Pratt (C. & N. W.): I would like to ask Mr. Haig if his road has used any flange lubricators on trailer wheels, and with what results? Also if they have had a large enough number of their engines equipped with flange lubricators to indicate whether lubrication on the flange of the rail would decrease the flange wear on other wheels of the engine or train? Also whether there has been any slipping of the engine on curves resulting from this flange lubrication?

G. A. Hancock (St. L. & S. F.): On the Frisco system we have at present about 50 lubricators. We use crude oil asphaltum as a base. What the saving is on the flange of the tender wheels I am unable to say, but we are getting very good results from it on the engines.

Angus Sinclair: A long time ago I was running a locomotive on a very crooked road. I was responsible for the condition

At the completion of this test move valve *A* to position No. 1 and open cock 9.

Test No. 5.—Packing-ring Leakage Test for Both Westinghouse and New York Triples.

Release Test, Sec. 1.—Commencing with cocks 1, 2, 3 and 9 open, all other numbered cocks closed, valve *A* in position No. 1, valve *B* in position No. 3 and the auxiliary reservoir charged to 80 lbs., proceed as follows:

Place the valve *A* in position No. 7 until the brake-pipe pressure is reduced 15 lbs., then return to position No. 3 (lap). Place valve *J* in position No. 2, lever *D* in notch No. 1 and valve *B* in position No. 1; close cocks 2 and 3 and move valve *A* to position No. 2. If the discharge does not occur promptly from the vent port of valve *B*, advance valve *J* from position to position until the discharge begins, then note the rate of increase of pressure on the auxiliary reservoir gage, which must not exceed 5 lbs. in 30 seconds.

During this test there must be a steady exhaust of air from the vent port of valve *B* to insure the proper differential being maintained on the triple-valve piston. If, in making this test, the triple valve for the 8-in. cylinder releases or indicates excessive ring leakage, make another test, beginning with moving handle *R* to the right after making the proper brake-pipe reduction and before starting to increase the brake-pipe pressure. Immediately after the test is completed, handle *R* should return to its normal left position.

Should it occur that the friction of the triple valves for the 10-in. brake cylinder is so low as to continue to permit the triple to release, the reduction for the application may be changed from 15 to 10 lbs. When this is done, special attention should be given to determining if the graduating valve is right, as it must be, to permit an accurate test.

At the completion of this test place valve *B* in position No. 3, open cocks 2 and 3 and place valve *A* in position No. 1.

Test No. 6, Sec. 2.—Friction Test. Release Test for Both Westinghouse and New York Valves.

Commencing the test with cocks 1, 2, 3 and 9 open and all other numbered cocks closed, valve *A* in position No. 1, valve *B* in position No. 3, auxiliary reservoir charged to 80 lbs.

Place lever *D* in notch 3 for all triple valves undergoing the test; proceed as follows:

Place valve *A* in position No. 7 until the brake-pipe pressure is reduced 10 lbs., then return it to position No. 3. Place valve *J* in position No. 1, valve *B* in position No. 1, and move valve *A* to position No. 2. Under these circumstances the triple valve should release. A failure to release should be accompanied by a discharge at the vent port of valve *B*, which indicates that the frictional resistance to the movement of the packing ring and slide valve is excessive.

If the triple valve does not release and valve *B* fails to open its exhaust, leakage is occurring from the brake pipe, which will necessitate advancing valve *J* from position to position, remaining in each position 30 seconds, until the triple valve releases or the exhaust in valve *B* opens.

At the completion of the test place valve *B* in position No. 3 and valve *A* in position No. 1.

Test No. 7, Sec. A.—Service-port Capacity Test for Westinghouse Triple Valves and Quick-service New York Triple Valves.

Commencing with cocks 1, 2, 3, 4 and 9 open, valve *A* in position No. 1, valve *B* in position No. 3, place valve *C* in position required for the triple valve under test, as indicated:

Notch No. 1.—For 8-in. triple valves.

Notch No. 2.—For 10-in. triple valves.

During this test the brake-pipe pressure should not drop, except that in the case of the quick-service triple valves there will, of necessity, be a slight drop, which must not exceed 2 lbs.

Place valve *B* in position No. 2 and move valve *A* to position No. 3, open cock 7 until brake-pipe and auxiliary-reservoir pressures are reduced to 50 lbs., then close cock 7. Move valve *B* to position No. 3 and open combination cock 6 and quick-opening valve, leaving it open 3 seconds. This test should not produce quick action. If it does, it indicates a restriction in the service port, or a weak or graduating spring.

SEC. B.—Duplicate the tests specified under Sec. A, placing the wheel of valve *C* in the position as indicated.

Notch No. 3.—For 8-in. triple valves.

Notch No. 5.—For 10-inch triple valves, excepting Westinghouse non-quick service, with which use notch 7.

This should result in the triple valve moving to emergency position. Failure to do so indicates too close a fit of the emergency piston.

At the completion of the test close cock 4 and combination cock 6 and quick-opening valve, move valve *A* to position No. 1.

Test No. 7, Sec. A.—Service-port Capacity Test for New York Non-quick Service Triple Valves.

Commencing with cocks 1, 2, 3, 4, and 9 open, valve *A* in position No. 1, valve *B* in position No. 3, place valve *C* in position required for the triple valve under test, as indicated.

Notch No. 1.—For 8-in. triple valves.

Notch No. 2.—For 10-in. triple valves.

Place valve *B* in position No. 2 and move valve *A* to position No. 3. Open cock 7 until brake pipe and auxiliary reservoir pressure are reduced to 50 lbs., then move valve *B* to position No. 3 and open cock 6 quickly.

NOTE.—During this test the triple valve should move to service position, the brake-pipe pressure must not drop and there must be no discharge of air from the vent ports.

Should the triple valve move to emergency position, it indicates a restriction in the service ports or a weak vent-valve spring.

SEC. B.—Duplicate the test specified under Sec. A, placing the wheel of valve C in the position as indicated for the triple valve under test.

Notch No. 3.—For 8-in. triple valves.

Notch No. 5.—For 10-in. triple valves.

This should result in the triple valve moving to emergency position, causing a strong blast of air from the vent posts and a brake-pipe reduction of at least 3 lbs. Failure to do so indicates a too loose fit of the vent-valve piston packing.

AIR-BRAKE HOSE COUPLINGS.

"To investigate and report on the question of standard dimension for air-brake hose couplings, gaskets and gages for new and worn hose couplings, and possibly signal-hose couplings."

This subject is referred to in paragraph 37, page 71, of the 1910 Proceedings:

"The report of committee on air-brake hose to the 1909 convention, with particular reference to 'standard dimensions of air-brake hose couplings and gaskets,' was received and referred to the committee on standards for further investigation and report. Inasmuch as this involved the design of the air-brake hose coupling, gasket and gages for both new and worn air-brake hose couplings, and should possibly be extended to signal-hose couplings, the committee felt that this matter should have been referred to the standing committee on train-brake and signal equipment, who have direct jurisdiction over the designing and interchangeability of these parts, and accordingly referred this subject to the committee on train-brake and signal equipment by letter, dated August 10, 1909, through the secretary of the association, who advised, under date of August 13, 1909, that the matter had been so referred."

The committee delegated this section of its work to a subcommittee, and its report, concurred in by all the members of your committee, follows:

Air-brake Hose Couplings and Packing Rings.

In its investigations the committee assumed that all air-brake hose couplings and packing rings now in use were supplied by the Westinghouse Air Brake Company and the New York Air Brake Company, or in accordance with their standard dimensions; and therefore the question of the best possible interchange of proposed standard and existing couplings and rings has received due consideration.

For the purpose of determining whether there are sufficient variations in the dimensions of various makes of couplings to justify the association in adopting a standard with a view of securing a more satisfactory interchange, more than 5,000 new and used couplings have been gaged and tested. This investigation revealed the fact that the dimensions of couplings which should be common to all makes vary appreciably in couplings supplied by different manufacturers.

Enlarged drawings have also been made showing the variations in the principal dimensions employed by the Westinghouse Air Brake Company and the New York Air Brake Company in the design of their respective coupling and packing ring. Outlines of the former and details of the latter show that there is a difference of 1/64 in. in the nominal height of a Westinghouse and a New York packing ring. Also, there is 3/64 in. difference in one of the principal dimensions of the couplings affecting the compression of the rings.

When two Westinghouse couplings, with Westinghouse packing ring in each, are coupled together, the nominal compression of each packing ring is 3/128 in. When the couplings are pulled apart (as is the case when cars are uncoupled without first parting the hose) the faces of the hose couplings are 1/32 in. apart.

By using a square nose tool and taking a coarser feed and heavier cut we have been able to reduce the time for finishing strap rod brasses from 3½ to 1½ hours for the smaller sizes, and from 5½ or six hours to 2½ hours for the larger ones. We used to turn our driving box brasses on a lathe, requiring 30 minutes. They were then laid off and fitted in to be planed, requiring 25 minutes, and about 25 minutes more for planing. This work is now done on a draw cut shaper, and the brasses are finished and ready to apply to the driving box in about 30 minutes for the smaller sizes and 45 minutes for the larger ones. It formerly required 12 hours to bore an 18-in. cylinder, this time being measured from the time the cylinder was taken off the floor until it was returned. This is now being done in nine hours, and the time will be reduced still more when the chuck shown in Fig. 2 has been installed. A is a section of the bed of the cylinder boring machine, and B is the base of the chuck, which may be used either for a cylinder bushing or a cylinder casting. The method of chucking one of the bushings is clearly shown in the illustration. To use this chuck with a cylinder it is necessary to remove the part C by taking out the bolt E and removing the block D and the pin F. The cylinder may then be placed on the base and adjusted to the proper position by manipulating the screws I and H. The casting may be securely clamped in position by means of two cross bars, one at each end of the cylinder, one end of the bars being held by eye bolts which fit on the pins F, and the other by T-bolts, which fit in slots G. The lugs L are bolted to the base of the machine by two bolts each.

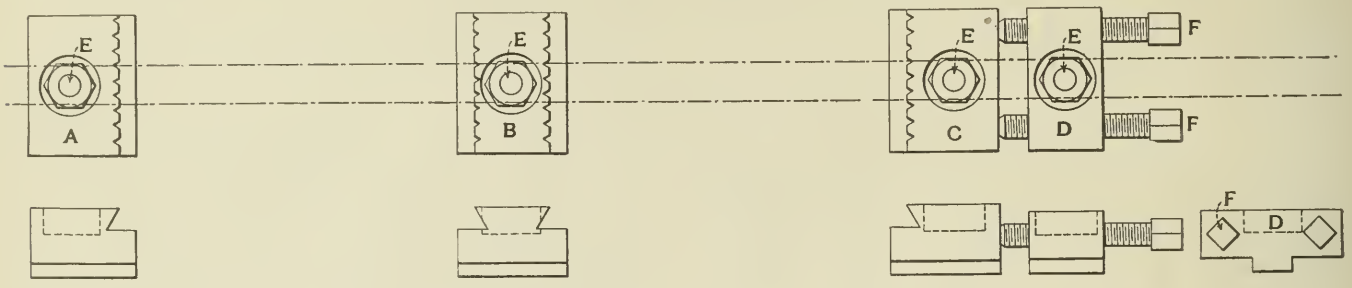


Fig. 3—Chuck for Milling Shoes and Wedges.

By using heavier tools and coarser feeds we have reduced the time for turning rod bushings on the lathe from one-half hour to 15 minutes. In the lathe it is the coarser feed and not the speed that counts. Formerly we turned our cylinder packing, of Stevenson style, on a lathe. By doing this on a boring mill we have been able to cut the time in half, turning out 30 rings in 10 hours. However, we are now installing Dunbar packing and are able to finish 10 rings in 10 hours.

We finish all shoes and wedges on a milling machine, machining both the inside and outside at one operation. One man will finish 40 of these in 10 hours. The chuck for milling shoes and wedges is shown in Fig. 3. The blocks A, B, C and D are made of cast steel and have tongues which fit in the grooves in the table of the milling machine. The blocks are held in place by the bolts E. The shoes are placed between the upper parts of the blocks and these are then forced tightly against the ends of the shoes by screwing up the set screws F in the tail block D. The shoes are milled at the top and sides in one operation. By changing the clamping device and forcing the cut for boring out driving boxes, we have been able to decrease the average time of boring these boxes and facing the hubs for from 6-in. to 9-in. journals from 1½ hours to 50 minutes. Five-inch strap rod brasses, which we formerly bored and faced on a lathe, requiring about 1½ hours, we are now able to finish on the boring mill in one hour. Formerly it was the practice to have two men on double-head bolt cutters. One man now attends to both heads, and we have increased the speeds for a ½-in. bolt from 122 to 200 r. p. m.; for a ⅝-in. bolt from 74 to 180 r. p. m.; for a ¾-in. bolt, 70 to 150 r. p. m., and for ⅞-in. and 1-in. bolts, 68 to 122 r. p. m. We also use high speed steel dies. One man has turned a thread for 2½ inches on ⅞-in. bolts at the rate of 160 in 30 minutes. Note that we are cutting the 1-in. bolts at the same rate formerly used for ½-in. bolts.

ERECTING SHOP.

The erecting shop was so crowded that we did not have adequate space to assemble brake work, spring rigging, pistons, crossheads and rocker boxes. Two pits were closed and used for this work, leaving 13 working pits. We were not using specialists on all work except steam pipe and brake rigging. One machinist and two apprentices remove all steam chests, fit all valve yokes, lay off new valve rods, fit new valves and valve strips and rings, face all valves and apply steam chests. One machinist and one apprentice do all the guide work. One machinist, an apprentice and a helper attend to the crossheads, pistons, cylinder packing and glands. Two handy men and one helper overhaul and apply all brake rigging and spring rigging, reaming all holes necessary for this work. Two machinists and two apprentices attend to the eccentrics and eccentric straps, apply the link motion and set the valves. One machinist and one apprentice remove, overhaul and apply the reverse levers and quadrants. One machinist and one helper face all frame jaws and apply the binders. One machinist and one apprentice lay off all shoes and wedges, fit up new ones and apply liners to the old ones. Three handy men and three helpers do the work on steam pipes, dry pipe, stand pipe and throttle rigging. One machinist is employed for cab work. A foreman and 12 helpers deliver all material, and when not engaged in this work, clean the shop. Each man in the erecting shop has a work box on a two-wheel truck, thus saving considerable time. Air hammers have been installed for doing all chipping.

We cast the brass on the shoe and wedge fits of both steel and cast iron driving boxes and also on the hubs. The boxes are not heated before the brass is poured. The arrangement for preparing the driving boxes for pouring the brass is shown in Fig. 4. The block D is placed in the box and the ring E, which forms the hub liner and which is 1 in. wide, ⅜ in. thick and of a diameter to suit the box, is placed in position, the block D projecting ½ in. above it, as shown in the illustration. The crown of the box is dove-tailed to hold the brass securely, the grooves H H crossing in the center and extending to within ¼ in. of the flange. Strips G are then clamped at the ends and the joints are luted with clay or asbestos. Attention is directed to our method of reclaning old driving boxes. Before we started applying the brass liners on the shoe and wedge fit we would scrap the box when it had been planed down to a certain limit. Now we plane the box to the limit and then pour a brass liner on it and use it again. This refers to cast iron boxes. The practice of casting liners on the steel driving boxes instead of riveting saves about $1.50 in labor on each box. We are welding our broken frames with crude oil without removing them from the engine.

We have a meeting of all of the foremen once a week, at which they are expected to report progress on the work in their departments for engines which are in the erecting shop and to state when they will be able to complete the work on each one of them. We then set a time at which each engine must be

HYPOTHETICAL CASE NO. 1.

Let it be assumed that on tramming to the valve stem with the main crank on the dead centers, the following irregularities in the lead are noticed for the engine under consideration. The dots on the diagrams represent the prick punch marks *F* and *F'* (Fig. 8) on the valve stem, while the crosses represent the irregularities in the lead when trammed to the valve stem (Fig. 12). The first procedure will be to divide the error between the forward and backward motions, as follows:

Error in forward motion—

Front, 3/8 in. — 1/4 in. lead = 1/8 in. error.
Back, 1/4 in. lead — 1/8 in. = 1/8 in. error.

} To square the lead the valve must be moved 1/8 in. ahead.

Error in backward motion—

Front, 7/16 in. — 1/4 in. lead = 3/16 in. error.
Back, 1/4 in. lead — 1/16 in. = 3/16 in. error.

} To square the lead the valve must be moved 3/16 in. ahead.

As the errors in the two motions occur in the same direction, it follows that the greater one partially neutralizes the effect of the lesser, and that the combined or average error will be the difference between the two; viz.: 3/16 in. minus 1/8 in. equals 1/16 in. average error.

To divide an average error of 1/16 in. equally about a central point, it will be necessary to move the valve one-half this amount, or 1/32 in. (in this case 1/32 in. back in forward motion). The eccentric rod must be shortened 1/16 in. (in the proportion of two to one) to move the valve 1/32 in. When this has been done the valve stem points will tram as shown in Fig. 13.

The errors in forward and backward motion have thus been equalized, and it remains only to square the lead, front and back, for both motions. The valve as now standing is 5/32 in. too far back to equalize the lead, viz.:

13/32 in. — 1/4 in. lead = 5/32 in. error front.
1/4 in. lead — 3/32 in. = 5/32 in. error back.

As the influene of the radius rod is direct, it follows that by lengthening this rod the amount required (5/32 in.) the valve will be squared, and can be trammed to the dimensions shown in Fig. 14. These dimensions are the ones required by the specification. The valve has thus been squared and the errors corrected by the following changes: Eccentric rod shortened 1/16 in. Link radius rod lengthened 5/32 in. A final trial of the valve and cut-off, etc., can now be made in the previously described manner.

HYPOTHETICAL CASE NO. 2.

Let it be assumed that on tramming for lead, results are obtained as represented by Fig. 15. Divide the error between the forward and backward motions, as follows:

Error in forward motion—

Front, 7/16 in. — 1/4 in. lead = 3/16 in. error.
Back, 1/4 in. lead — 1/16 in. = 3/16 in. error.

} To square the lead the valve must be moved 3/16 in. ahead.

Error in backward motion—

Front, 1/4 in. lead — 3/16 in. = 1/16 in. error.
Back, 5/16 in. — 1/4 in. lead = 1/16 in. error.

} To square the lead the valve must be moved 1/16 in. back.

As the errors in the two motions occur in opposite directions, it follows that they augment each other, and that the combined or average error will be the sum of the two, viz.: 3/16 in. plus 1/16 in. equals 1/4 in. average error. To divide this error equally about a central point, it will be necessary to move the valve one-half the amount, or 1/8 in. (in this case 1/8 in. ahead in forward motion). The eccentric rod must be lengthened 1/4 in. (in the proportion of two to one) to move the valve 1/8 in. When this has been done, the valve stem will tram as shown in Fig. 16.

The errors in forward and backward motion have thus been equalized, and it remains only to square the lead front and back for both motions. The valve as now standing is 1/16 in. too far back to equalize the lead, viz.:

5/16 in. — 1/4 in. lead = 1/16 in. error, front.
1/4 in. lead — 3/16 in. = 1/16 in. error, back.

To move the valve ahead 1/16 in. the link radius rod must be lengthened 1/16 in. and the lead will then be squared. When trammed for lead, the results will be as shown by Fig. 14. These dimensions are the ones required by the specification.

The lead has been squared and the errors have been corrected by the following changes: Eccentric rod lengthened 1/4 in. Link radius rod lengthened 1/16 in.

Trial of the valve travel and cut-off, etc., can now be made in the manner previously described. From the above it is evident, that the errors in forward and backward motion are equalized by changing the length of the eccentric rod; and the lead is then squared by changing the length of the radius rod. Theoretically the radius rod should not be changed; but the amount necessary is so slight, that it makes practically no difference in the movement of the valve.

With outside admission piston valves the arrangement of the gear is the same as that used with outside admission slide valves, and the method of setting is the same. The line and line positions of the valve, however, must be observed through peep holes, as in the case of the inside admission piston valve.

THE Y. M. C. A. PLAN OF INSTRUCTION FOR APPRENTICES AND SHOP EMPLOYEES.*

BY F. M. M. RICHARDSON.

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In September, 1906, the Young Men's Christian Association undertook to conduct the apprenticeship school of the Westinghouse Air Brake Company at Wilmerding, Pa., on company time. Conducting apprenticeship schools for corporations was not a new experience to the Young Men's Christian Association, but it was new to conduct one during the working hours. It meant that the apprentice received the same rate of pay for attending classes as for some productive work in the shop, and that he received special instruction better fitting him for the trade in which he was indentured. In the last five years considerable progress has been made in the development of such schools and the day plan has been successfully adopted by a number of industrial and railway corporations.

The first class of railway shop apprentices under the Association plan was opened in November, 1909, at the Pine Bluff shops of the St. Louis Southwestern. This was closely followed by the opening of classes on the Delaware, Lackawanna & Western at Scranton in January, 1910, and later with classes at East Buffalo and Kingsland, thus placing all the apprentices of that road under instruction. On the St. Louis Southwestern the work begun at Pine Bluff was extended to Tyler, Texas, at the end of the first year, thus including all apprentices on that system.

The plan of operation is based on the idea of uniform instruction over the lines of any railway system and is a part of the system educational work of the railway department of the International Committee of Young Men's Christian Associations. Where a Railway Association is located at a shop point the work is done in co-operation with that Association. The absence of such an Association does not, however, prevent the establishment of a school at any point, as the instruction is generally conducted in a room provided by the company on the shop grounds.

In considering a modern plan for the instruction of apprentices the first two questions to be met are those of cost and allowing time out of shop. The latter question is involved in the cost but its significance is in its relation to the working conditions prevailing in the shop, and it is summed up in the query, "To what extent would the output of the shop he affected by releasing the apprentice for a number of hours' instruction each week?" The question can best be answered by those who have tried the day plan of instruction and have found that the increased efficiency of the apprentice due to instruction more than offsets the loss of the several hours spent in the class room each week. There is still a tendency on the part of some to believe that while this may be true of the shop with a large number of apprentices, it cannot apply to the shop where but a few are employed and where the work of each apprentice counts as a definite part of the shop output. However, with a flexible plan like that of the Association the local conditions may be met so as to give to the shop a practicable working scheme for the instruction of the apprentices employed.

In one of the shops where the Association plan is in operation there are but fourteen apprentices, divided among four trades. Each apprentice attends one class session in one three-hour period, but the apprentices are divided into two classes, so that by this arrangement not more than half the apprentices in each department are out of shop at any one time. In another shop the group is divided into four classes, and thus only one-fourth of the apprentices in each department are out of the shop for one period of instruction once each week. An arrangement of this kind brings together the apprentices of the different trades for instruction in the same class period, but the class is small, making a splendid condition for individual instruction, which is the key to effective results. The instructor is thus permitted to meet

*Entered in the competition on the instruction of workmen and apprentices which closed April 15, 1911.